EXP 2

IPYWidgets-Based Shoping Cart and billing System

AIM:

To set up the Python environment by installing and checking the ipywidgets and jupyterlab_widgets libraries, and to use Jupyter widgets to build an interactive order form with input fields, buttons and tabs that calculates a bill.

ALGORITHM:

1. Import ipywidgets and the display and clear_output functions from IPython.display.
2. Create the text input widgets for Name, Email and Address.
3. Create the Product dropdown and the Quantity, Discount and Tax sliders.
4. Create the Gift Wrap checkbox, the Colour picker and the Payment toggle buttons (Cash, UPI, Card).
5. Create the Add to Cart, Checkout and Clear buttons and an Output widget.
6. Store the product prices in a dictionary and create an empty list for the cart.
7. Define add_item() to add the selected product and quantity to the cart and show the details.
8. Define bill() to check that the cart is not empty, calculate the total, discount, tax and final amount, and print the bill.
9. Define clear_cart() to empty the cart and show a message.
10. Connect each button to its function using on_click().
11. Arrange the widgets in three tabs (Customer Details, Product Selection, Order Summary) and display them.
12. Install or upgrade jupyterlab_widgets and ipywidgets using %pip so that the widgets are displayed correctly.

CODE:

```
import ipywidgets as widgets
from IPython.display import display, clear_output
```

```
name = widgets.Text(description="Name:")
email = widgets.Text(description="Email:")
address = widgets.Textarea(description="Address:")
```

```
product = widgets.Dropdown(
    options=["Laptop", "Mobile", "Headphones"],
    description="Product:"
)
quantity = widgets.IntSlider(min=1, max=10, value=1, description="Qty:")
discount = widgets.IntSlider(min=0, max=30, step=5, description="Discount:")
tax = widgets.FloatSlider(min=0, max=30, value=18, description="Tax:")
```

```
gift = widgets.Checkbox(description="Gift Wrap")
color = widgets.ColorPicker(description="Color:")
payment = widgets.ToggleButtons(
    options=["Cash", "UPI", "Card"],
```

```
    description="Payment:"
)
```

```
add = widgets.Button(description="Add to Cart", button_style="success")
checkout = widgets.Button(description="Checkout", button_style="primary")
clear = widgets.Button(description="Clear", button_style="danger")
output = widgets.Output()
```

```
prices = {
    "Laptop": 50000,
    "Mobile": 25000,
    "Headphones": 2000
}
cart = []
```

```
def add_item(b):
    cart.append((product.value, quantity.value))
    with output:
        clear_output()
        print("Added to Cart!")
        print("----------------------")
        print("Product:", product.value)
        print("Quantity:", quantity.value)
        print("Price: $", prices[product.value])


# Checkout
def bill(b):
    with output:
        clear_output()

        if not cart:
            print("Cart is empty!")
            return

        total = sum(prices[p] * q for p, q in cart)

        discount_amount = total * discount.value / 100
        after_discount = total - discount_amount

        tax_amount = after_discount * tax.value / 100
        final = after_discount + tax_amount

        print("Customer:", name.value)
        print("Email:", email.value)
        print("Address:", address.value)
        print("----------------------")

        for p, q in cart:
            print(p, "x", q)

        print("----------------------")
        print("Total: $", total)
        print("Discount:", discount.value, "%")
        print("Tax:", tax.value, "%")
        print("Final Amount: $", final)
        print("Payment:", payment.value)
        print("Gift Wrap:", gift.value)
```

```
        print("Order Confirmed!")
def clear_cart(b):
    cart.clear()
    with output:
        clear_output()
        print("Cart Cleared!")


# Button Events
add.on_click(add_item)
checkout.on_click(bill)
clear.on_click(clear_cart)
```

```
tabs = widgets.Tab([
    widgets.VBox([
        widgets.HBox([name, email]),
        address,
    ]),

    widgets.VBox([
        product,
        quantity,
        discount,
        tax,
        payment,
        gift,
        color,
        widgets.HBox([add, clear])
    ]),

    widgets.VBox([
        checkout,
        output
    ])
])

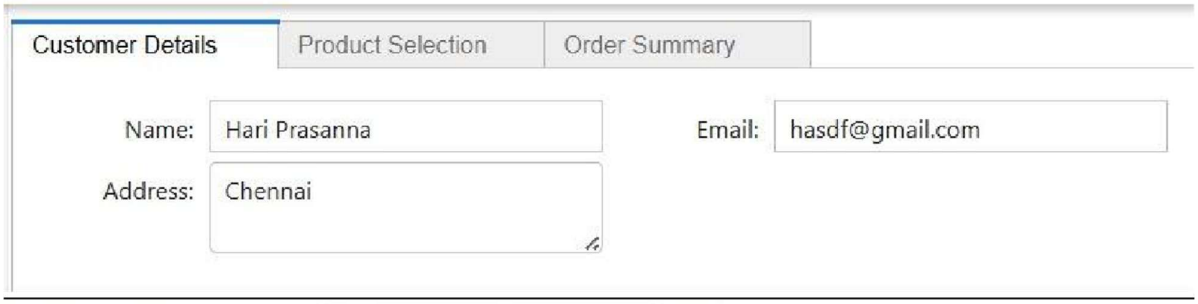
# Tab Names
tabs.set_title(0, "Customer Details")
tabs.set_title(1, "Product Selection")
tabs.set_title(2, "Order Summary")

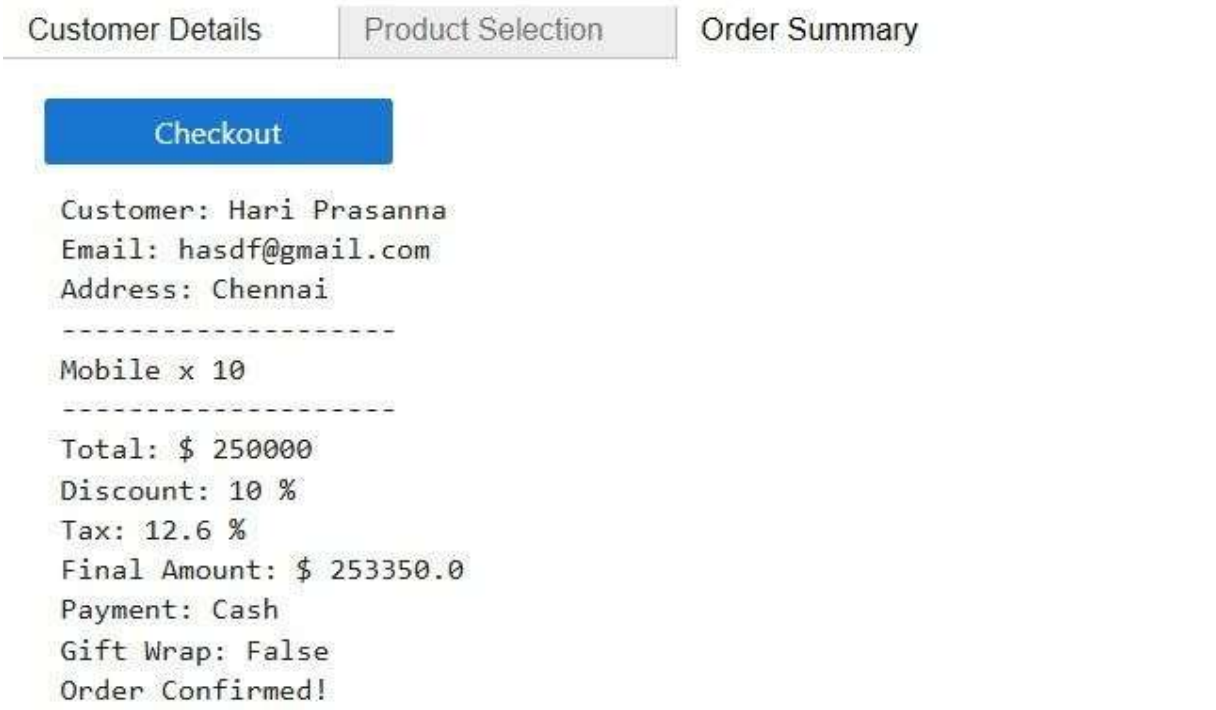
display(tabs)
```

```
%pip install --upgrade jupyterlab_widgets
```

```
%pip install --upgrade ipywidgets
```

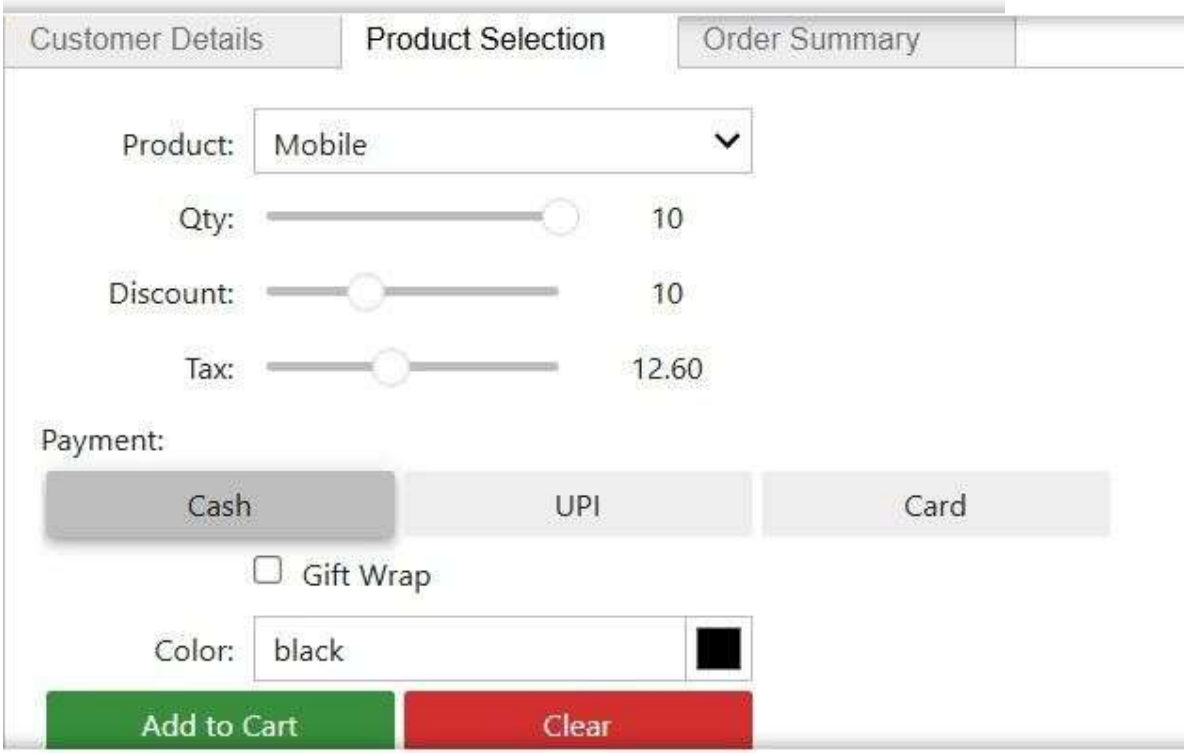
OUTPUT:


Customer Details
Product Selection
Order Summary
Name:
Hari Prasanna
Email:
hasdf@gmail.com
Address:
Chennai


Customer Details
Product Selection
Order Summary
Checkout
Customer: Hari Prasanna
Email: hasdf@gmail.com
Address: Chennai

Mobile x 10

Total: $ 250000
Discount: 10 %
Tax: 12.6 %
Final Amount: $ 253350.0
Payment: Cash
Gift Wrap: False
Order Confirmed!


Customer Details
Product Selection
Order Summary
Product:
Mobile
Qty:
10
Discount:
10
Tax:
12.60
Payment:
Cash
UPI
Card
Gift Wrap
Color:
black
Add to Cart
Clear

RESULT:

Thus, the Python environment was set up with the ipywidgets and jupyterlab_widgets libraries, and an interactive three-tab order form was created in Jupyter Notebook. The Add to Cart, Checkout and Clear buttons are linked to functions that manage the cart and calculate the final bill with discount and tax.